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JAVA PROGRAMMING

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192525404RB

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         int n=3;
5         int fact=1;
6         for(int i=1;i<=n;i++){
7             fact=fact*i;
8         }
9         System.out.println("factorial is:"+fact);
10    }
11 }
```

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OUTPUT

```
factorial of is:6
```

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QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class first
2 {
3     public static void main(String[] args)
4     {
5         int n=7;
6         boolean isPrime = true;
7         if(n<=1){
8             isPrime = false;
9         }
10        else{
11            for(int i=2;i<n;i++){
12                if(n%i==0){
13                    isPrime = false;
14                    break;
15                }
16            }
17        }
18        if(isPrime){
19            System.out.println(n+"is a prime number");
20        }
21        else{
```

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```
16     }
17     }
18     if(isPrime){
19         System.out.println(n+"is a prime number");
20     }
21     else{
22         System.out.println(n+"is not a prime number");
23     }
24 }
25 }
```

OUTPUT

```
7is a prime number
```

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QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         int year=2024;
5         if((year%400==0)|| (year%4==0 && year%100!=0)){
6             System.out.println("leap year");
7         }
8         else{
9             System.out.println("not leap year");
10        }
11    }
12 }
```

INPUT

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OUTPUT

leap year

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QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG
Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class armstrong
2 {
3     public static void main(String[] args)
4     {
5         int num=153;
6         int d1=num/100;
7         int d2=(num/10)%10;
8         int d3=num%10;
9         int sumOfCubes=(d1*d1*d1)+(d2*d2*d2)+(d3*d3*d3);
10        Boolean isArmstrong=(num==sumOfCubes);
11        System.out.println("Is Armstrong?" + isArmstrong);
12    }
13 }
```

INPUT

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OUTPUT

Is Armstrong?true

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QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI
Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 public class main
2 {
3     public static void main(String[] args)
4     {
5         int n=10;
6         int first=0,second=1;
7         System.out.print("fibonacci series:");
8         for(int i=1;i<=n;i++){
9             int next=first+second;
10            first=second;
11            second=next;
12            System.out.println(" "+next);
13        }
14    }
15 }
16 }
```

INPUT

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OUTPUT

```
fibonacci series: 1
2
3
5
8
13
21
34
55
89
```

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QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class main{
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in);
6         System.out.print("enter a number:");
7         int num=sc.nextInt();
8         int dup=num;
9         int rev=0;
10        while(num>0){
11
12
13            rev=rev*10+num%10;
14
15            num=num/10;
16        }
17        if(dup==rev){
18            System.out.println("palindrome");
19        }
20        else{
21            System.out.println("not palindrome");
22        }
23    }
24 }
```

INPUT

121

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```
18         System.out.println("palindrome");
19     }
20     else{
21         System.out.println("not palindrome");
22     }
23     System.out.println("reversed number = "+rev);
24 }
25 }
```

OUTPUT

```
enter a number:palindrome
reversed number = 121
```

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QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 class main{
3     public static void main(String[] args)
4     {
5         Scanner sc=new Scanner(System.in);
6         int num=sc.nextInt();
7         int sum=0;
8         while(num!=0){
9             sum=sum+num%10;
10            num=num/10;
11        }
12        System.out.println("sum of digits:"+sum);
13    }
14 }
```

INPUT

14

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14 }

OUTPUT

sum of digits:5

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QUESTIONS 8 / 20 completed

Shapes-1

SHAPES-1

Write a Java program to implement a basic shape hierarchy using inheritance. Create a parent class called Shape with attributes such as color and area. Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have

PROGRAM EDITOR

Java

Run

Save

```
15  if(arr[i]>arr[j]){
16      int temp=arr[i];
17      arr[i]=arr[j];
18      arr[j]=temp;
19  }
20  }
21  }
22  int i;
23  int count=1;
24  for(i=0;i<n;i++)
25  {
26      if(i == n-1 || arr[i] != arr[i+1]){
27          System.out.println(arr[i]+" = "+count);
28          count=1;
29      }
30      else{
31          count++;
32      }
33  }
34  }
35  }
```

INPUT

5
1
1
2
4
3

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Square, and Triangle that inherit from the Shape class and have

```
28  count=1;
29  }
30  else{
31      count++;
32  }
33  }
34  }
35  }
```

OUTPUT

1 = 2
2 = 1
3 = 1
4 = 1

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QUESTIONS 31 / 31 completed

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class lcm
2 {
3     public static void main(String[] args){
4         int num1=15;
5         int num2=20;
6         int a=num1;
7         int b=num2;
8         while(b!=0){
9             int temp=b;
10            b=a%b;
11            a=temp;
12        }
13        int gcd=a;
14        int lcm=(num1*num2)/gcd;
15        System.out.println("lcm:"+lcm);
16    }
17 }
```

INPUT

Your INPUT goes here!

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OUTPUT

```
lcm: 60
```

```
14     int lcm=(num1*num2)/gcd;
15     System.out.println("lcm:"+lcm);
16 }
17 }
```

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QUESTIONS 31 / 31 completed

GCD

GCD
Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 public class gcd
2 {
3     public static void main(String[] args)
4     {
5         int a=56;
6         int b=98;
7         while(b!=0){
8             int temp=b;
9             b=a%b;
10            a=temp;
11        }
12        System.out.println("gcd:"+a);
13    }
14 }
```

INPUT

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OUTPUT

gcd:14

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Nth Term Java Code

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QUESTIONS 8 / 20 completed

Games

GAMES

Write a Java program to implement a basic game system using inheritance. Create a parent class called Game with attributes such as title and genre. Then, create child classes such as ActionGame and PuzzleGame that inherit from the Game

PROGRAM EDITOR

Java

Run

Save

```
5 {
6 Scanner sc = new Scanner(System.in);
7 int n=sc.nextInt();
8 int arr[]=new int[n];
9 for(int i=0;i<n;i++){
10 arr[i]=sc.nextInt();
11 }
12 int i,j=0;
13 int count=0;
14 for(i=0;i<n;i++){
15 for(j=i+1;j<n;j++){
16 if(arr[i]==arr[j]){
17 System.out.println(arr[i]);
18 count++;
19 }
20 }
21 }
22 System.out.println("no of duplicants:"+count);
23 }
24 }
25 }
```

INPUT

3
10
5
10

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Nth Term Java Code

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ActionGame and PuzzleGame that inherit from the Game

OUTPUT

10
no of duplicants:1

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:
**

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         for(int i=1;i<=5;i++){
5             for(int j=1;j<=i;j++){
6                 System.out.print("** ");
7             }
8             System.out.println();
9         }
10    }
11 }
```

INPUT

Your INPUT goes here!

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OUTPUT

```
**
***
****
*****
```

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         for(int i=1;i<=5;i++){
5             for(int j=1;j<=i;j++){
6                 System.out.print("** ");
7             }
8             System.out.println();
9         }
10        for(int i=5;i>=1;i--){
11            for(int j=1;j<=i;j++){
12                System.out.print("** ");
13            }
14            System.out.println();
15        }
16    }
17 }
```

INPUT

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14 System.out.println();

15 }

16 }

17 }

OUTPUT

```
***
****
*****
****
***
**
*
```

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         int n=5;
5         for(int i=1;i<=n;i++){
6             for(int j=1;j<=n-i;j++){
7                 System.out.print(" ");
8             }
9
10
11         for(int k=1;k<=(2*i-1);k++){
12             System.out.print(" *");
13         }
14         System.out.println();
15     }
16 }
17 }
18 }
```

INPUT

Your INPUT goes here!

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14 System.out.println();

15 }

16 }

17 }

18 }

OUTPUT

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         int n=5;
5         for(int i=n;i>=1;i--){
6             for(int j=1;j<=n-i;j++){
7                 System.out.print(" ");
8             }
9         }
10        for(int k=1;k<=(2*i-1);k++){
11            System.out.print("*");
12        }
13        System.out.println();
14    }
15 }
16 }
17 }
18 }
```

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OUTPUT

**
*

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QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR

Java

Run

Save

```
1 class main{
2     public static void main(String[] args)
3     {
4         int rows=5;
5         int columns=5;
6         for(int i=1;i<=rows;i++){
7             for(int j=1;j<=columns;j++){
8                 if(i==1 || i==rows || j==1 || j==columns){
9                     System.out.print("**");
10                }
11            }
12            else {
13                System.out.print(" ");
14            }
15            System.out.println();
16        }
17    }
18 }
19
```

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```
14     }
15     System.out.println();
16 }
17 }
18 }
19
```

OUTPUT

```
*****
*  *
*  *
*  *
*****
```

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